1 Find the equation of a line perpendicular to the line $5x - 2y - 11 = 0$ passing through the point $(2, -3)$. [4]

---

2 Solve the simultaneous equations [5]

$2x + y = 7,$

$xy - x^2 = 4.$

---

3 Find the range of values of k, where $k > 0$, for which the line $y = kx - 2$ does not meet the curve $x^2 - 4y + 1 = 0$. [4]

---

4 The functions g and h are defined by $g: x \to \frac{1}{2x+1}$, $x \neq -\frac{1}{2}$ and h: $x \to x + 3$ respectively.

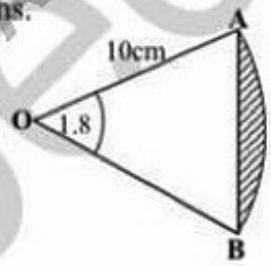
Find

(a) $\mathrm{hg}(x)$, [2]

(b) the values of x for which $\mathrm{hg}(x) = \mathrm{h}(x)$. [4]

---

5 In the diagram below, OAB is a sector of a circle with centre O, radius 10cm and angle AOB = 1.8 radians.



Find

(a) the perimeter of the shaded region, [3]

(b) the area of the shaded region. [2]

---

6 Prove the identity

$\tan\theta + \cot\theta \equiv \sec\theta \operatorname{cosec}\theta.$ [4]

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---

7 **(a)** Find the term independent of x in the expansion of $\left(3x + \frac{1}{x}\right)^{10}$. [4]

(b) Find the coefficient of x^2 in the expansion of $(1 + x)^7 (1 - 2x)^4$. [5]

---

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8 **(a)** Find $\frac{dy}{dx}$ for the equation $y = \frac{3x-5}{x^2-4x}$. [4]

(b) Two variables x and y are related by the equation $y = 2x^2 - 3x$. Obtain an expression for $\frac{dy}{dx}$ and find the approximate change in y as x increases from 6 to 6.02. [5]

---

9 Find all the angles between 0° and 360° which satisfy the equation

$|3\sin\theta - 1| = 2$. [5]

---

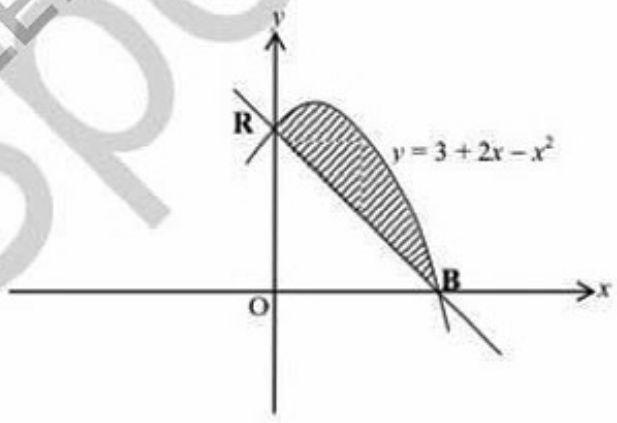
10 **(a)** A and B are points with position vectors $3i + j$ and $10i + j$ respectively. Find the position vector of a point which divides $\overrightarrow{AB}$ in the ratio 3:4. [4]

(b) The position vectors of P and Q are $ai + 2j$ and $3i + j$. Given that the angle between p and q is 45°, find two possible values of a. [5]

---

11 **(a)** Evaluate $\int_1^3 (3x-4)\,dx$. [2]

(b) The diagram below shows part of the curve $y = 3 + 2x - x^2$ which meets a straight line at B and R.



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Find

(i) the coordinates of B, [2]

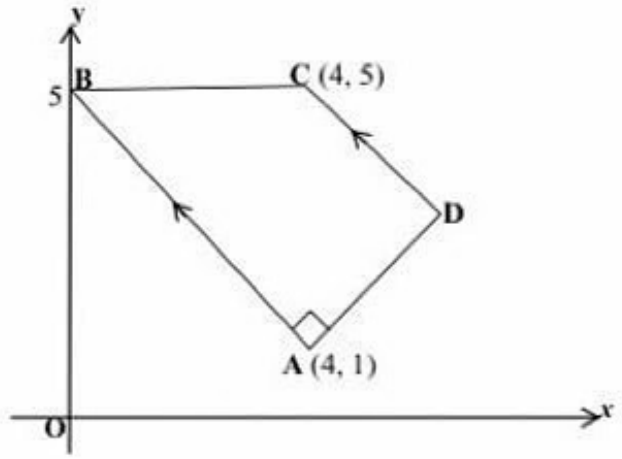
(ii) the equation of BR, [1]

(iii) the area of the shaded region. [5]

---

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12 In the diagram below, ABCD is a trapezium with A(4, 1) and C(4, 5). The point B lies on the y-axis and angle DAB = 90°.



Find

(a) the equation of line AB, [3]

(b) the equation of line AD, [3]

(c) the coordinates of D. [4]

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EXAMINATIONS COUNCIL OF ZAMBIA

Examination for School Certificate Ordinary Level

Additional Mathematics 4030/1

Paper 1

Specimen

Additional Materials:

Answer Booklet
Silent electronic calculator (Non programmable)

Time: 2 hours

Instructions to Candidates

Write your **name, centre number** and **candidate number** in the spaces on the Answer Booklet provided.

There are **12 questions** in this paper. Answer **all** the questions.

Write your answers in the **Answer Booklet provided**.

If you use more than one Answer Booklet, **fasten** the Answer Booklets together.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

Information for candidates

The number of marks is shown in brackets [ ] at the end of each question or part question.

The total number of marks for this paper is 80.

The use of a non programmable electronic calculator is expected, where appropriate.

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Cell phones are not allowed in the examination room.

You are reminded of the need for clear presentation in your answers.

Check the formulae overleaf.

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MATHEMATICS FORMULAE

1 ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial Theorem

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \ldots + \binom{n}{r}a^{n-r}b^r + \ldots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{(n-r)!r!}$

2 TRIGONOMETRY

Identities

$\text{Sin}^2 A + \cos^2 A = 1$

$\text{Sec}^2 A = 1 + \tan^2 A$

$\text{Cosec}^2 A = 1 + \cot^2 A$

Formulae for ΔABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$a^2 = b^2 + c^2 - 2bc \cos A$

$$\Delta = \frac{1}{2} bc \sin A$$

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